

Final Project

Forecasting UFO Sightings

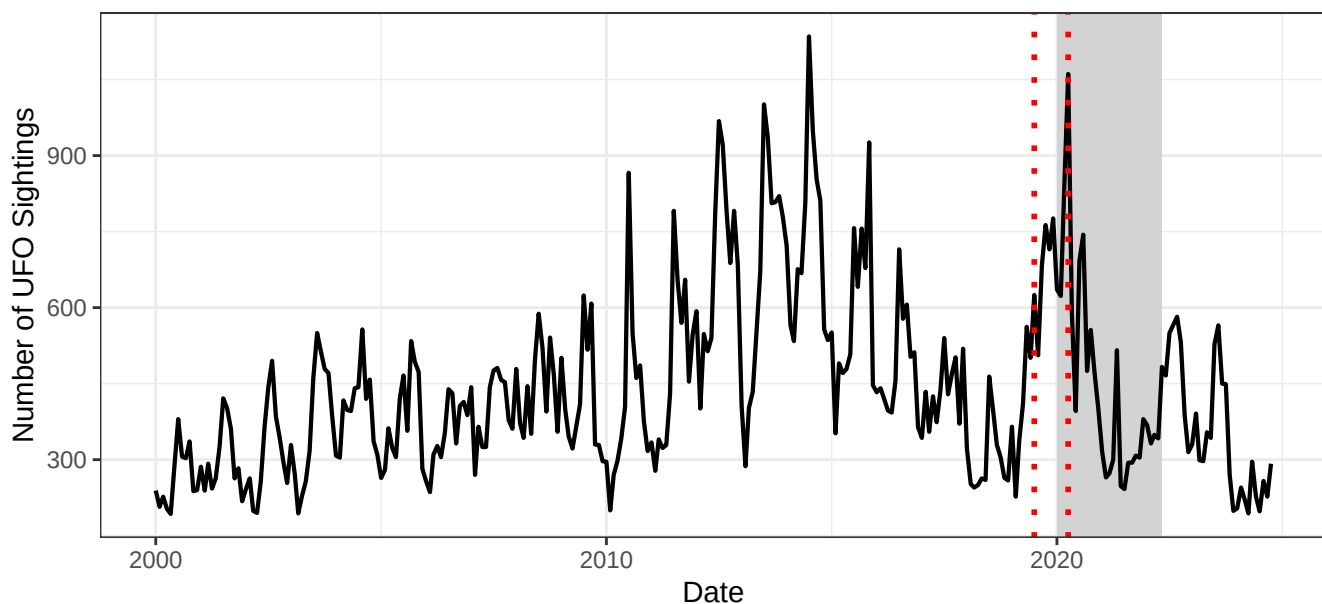
Kayley, Bruck, Grayson



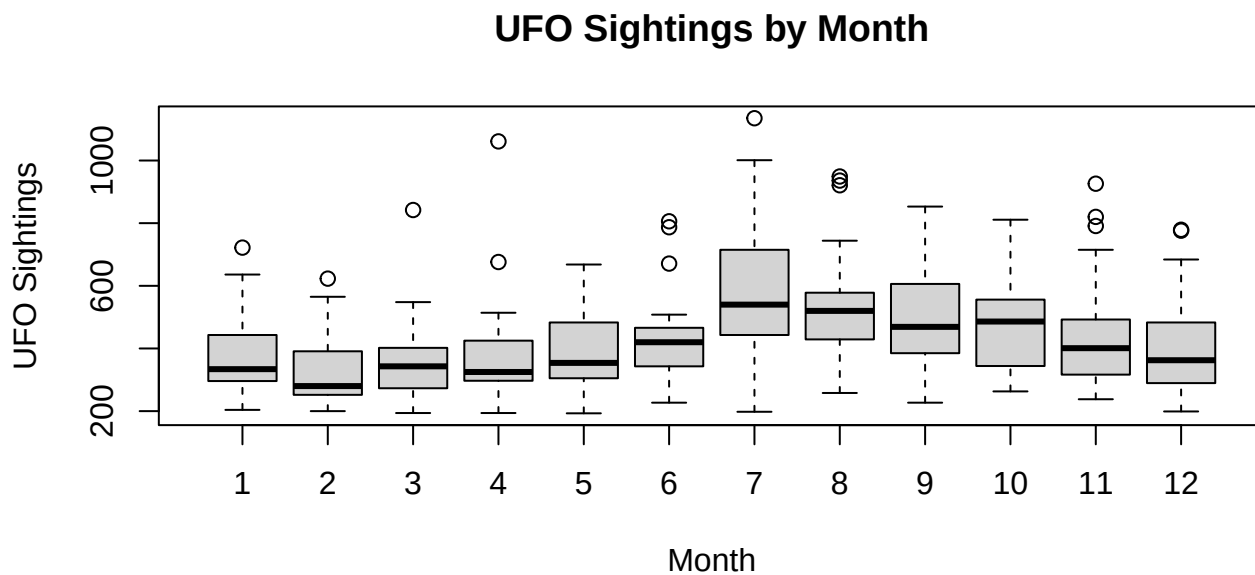
Introduction

Are we really alone in the universe? Since the beginning of recorded history, humans have been fascinated by unidentified objects in the sky. With limited scientific knowledge, early civilizations often attributed these sightings to gods or deities. As our understanding of science and the universe increased, many of these sightings could be explained. However, scientific progress never fully eliminated the idea that unidentified flying objects (UFOS) are caused by alien life forms. Even today, the internet is rife with posts and videos claiming to have proof of alien life. We found a dataset from the NUFORC website that quantified the number of UFO sightings worldwide per month. From initial exploration of the data, we developed our research question: Is there an identifiable pattern or trend in the number of UFO sightings per month and is the number of future sightings predictable using stochastic methods?

Monthly UFO Sightings Since 2000



For a quick first look at our data, we created a graph of the monthly sightings and plotted over time. The graph appeared to have a moderate seasonal component, so we examined the high and low values to see if we could identify a causal factor in the change in sightings. We initially thought that the covid-19 pandemic would cause an increase in the number of sightings, however after highlighting the corresponding date range we found that this was not true. We then started to investigate whether the spikes corresponded with media coverage of aliens or UFO sightings. We highlighted two events, the 2019 Area 51 Raid, and the pentagons 2020 declassification of two ufo videos, as events that possibly led to spikes in their respective months. However, not every spike could be traced back to a specific news story or event.



We also wanted to explore the possibility of a seasonal component that might correspond with daylight hours or season, or a yearly holiday. To visualize this, we created 12 boxplots, one for each month. We saw somewhat of a spike in July, but since the error bars were almost the same as every other month, we came to the conclusion that this difference we were seeing was either not significant or caused by something so small or unusual we would not be able to identify what exactly it was.

Methods

The NUFORC database contains alleged and sporadic reports dating back to 1864. Given that the NUFORC was founded in 1974, it's hard to take these historical reports seriously. We decided to ignore all data points before the year 2000 because after that point, the internet had wider general use and data is more likely to be accurate and comparable to modern reporting. Assuming that the spike in the trend around 2019 was caused by the Covid pandemic, we ignored all data after 2019 in hopes that we could forecast this spike.

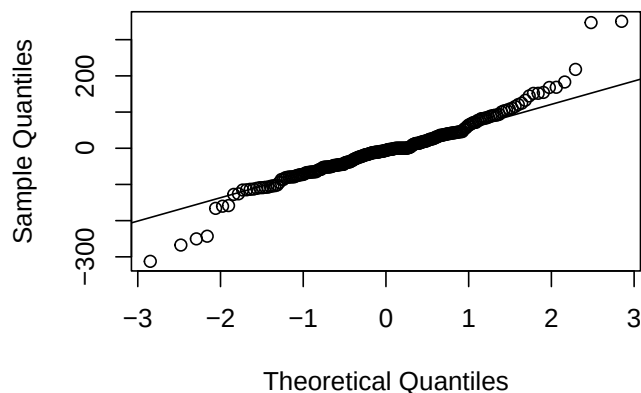
We decided to approach the task of forecasting our UFO sightings by using a seasonal ARIMA model where we allowed the Auto.arima function to estimate the most optimal parameters for the auto regressive, integrated, moving average, and seasonal components for the purpose of fitting and forecasting. Auto.arima's strength in forecasting time series is where a clear seasonal trend is able to be exploited. We attempted to use the strength of Auto.arima, however since there isn't any meaningful seasonality so the forecast was very poor.

We then tried to use a simpler forecasting tool in Holt-Winters, which is different from Auto.arima in how it uses the provided data to forecast. Unlike Auto.arima, Holt-Winters does not weigh more recent observations as more important and instead takes into account the overall trend of the data when making predictions. This resulted in a model that predicted a positive trend in sightings, since most of

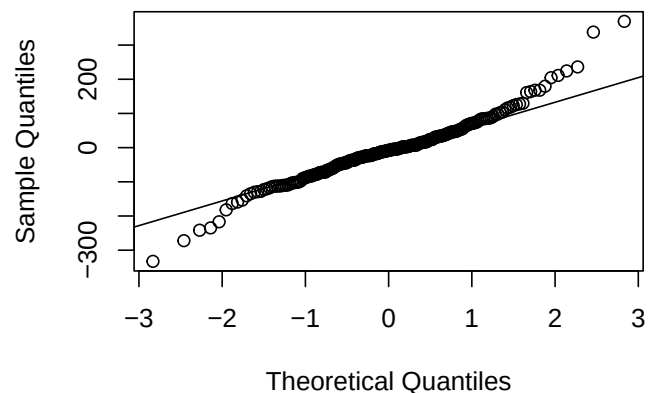
the data is increasing for a majority of the observation cycle. Even though Holt-Winters was also weak at predicting future values, it was slightly better than Auto.arima.

When assessing the assumptions of each model, it is important to understand the components that are needed in order for the model to produce valid, interpretable results. In this case, the 95% confidence interval of the forecast is dependent on normality because it uses a standard normal distribution in order to calculate the values of the “bounds” of the interval. The first assumptions of normality, which can be assessed by looking at the Q-Q plot of the residuals.

Normal Q-Q Plot for ARIMA Model

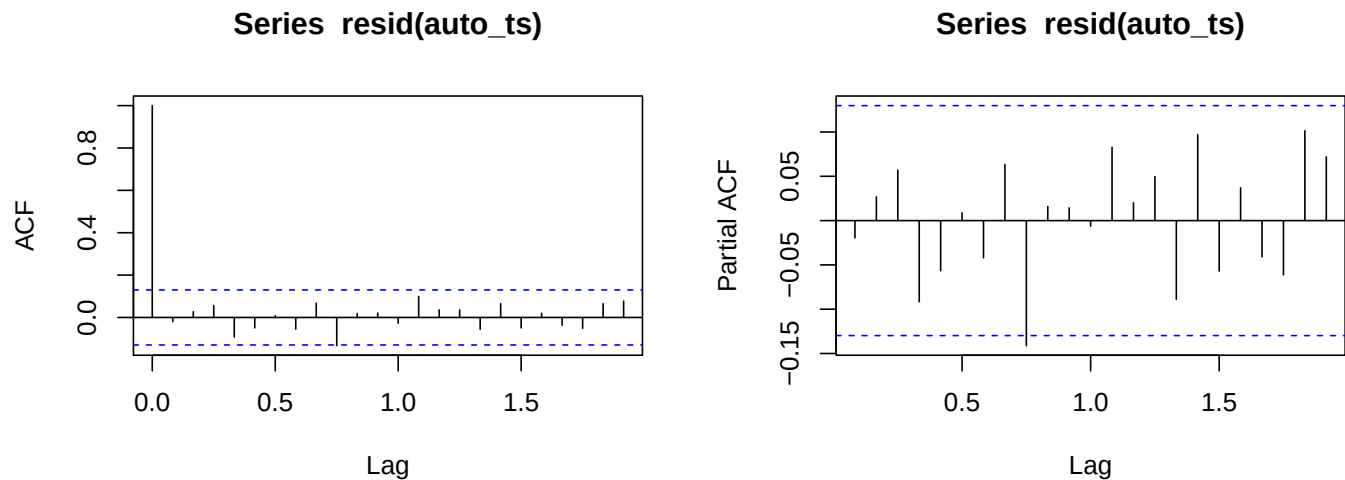


Normal Q-Q Plot for Holt-Winters Model

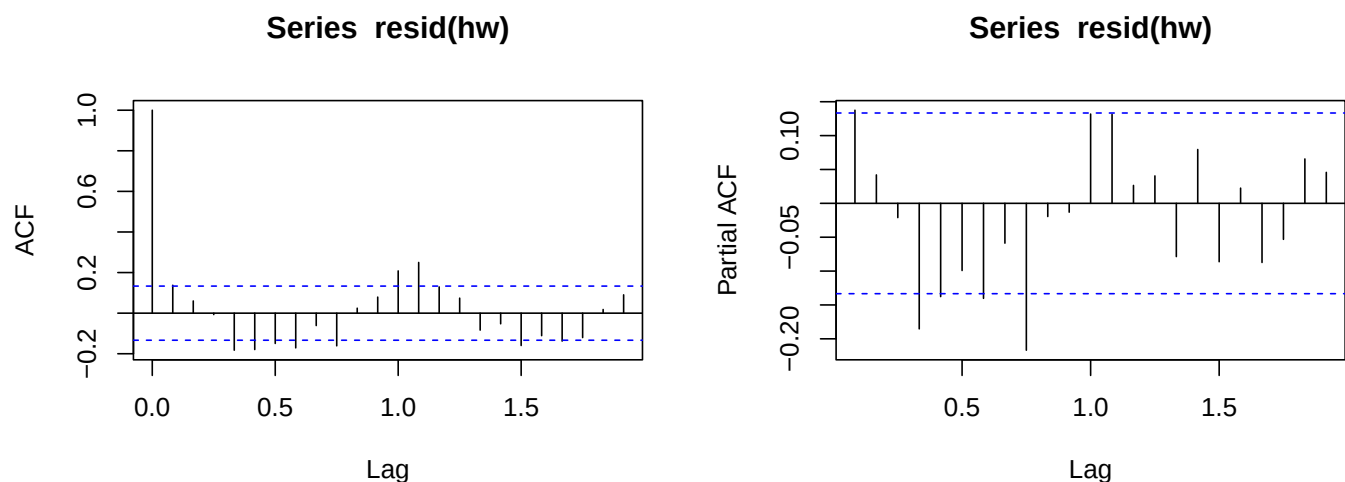


For Auto.arima there seems to be evidence against the assumption of normality as before the two standard deviation mark there seems to be some evidence of a heavy tailed distribution as the points skew at the tails. For Holt-Winters, there does not seem to be evidence against the assumption of normality prior to the 2 standard deviation mark as there is minimal skewing from the 45-degree reference line. However, due to the Central Limit Theorem, we expect the sampling distribution of the forecasts to be approximately normally distributed.

Assessing independence is important because if there is residual autocorrelation identified in the Auto-correlation Function (ACF) and Partial Autocorrelation Function (PACF) graphs then the model “fit” to the data is not capturing all the relationships within the data which can lead to poor forecasting ability of the model.

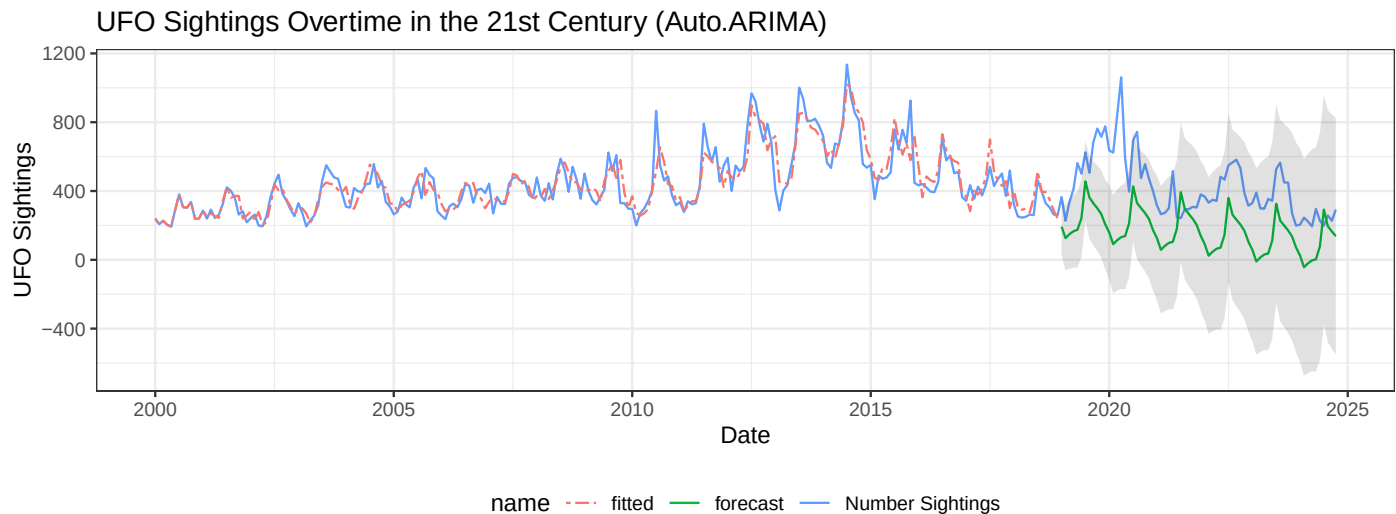


When looking at the ACF and PACF plots of the Auto.arima model there does not seem to be any remaining residual autocorrelation, such that there is no evidence against the assumption of independence for the Auto.arima model.

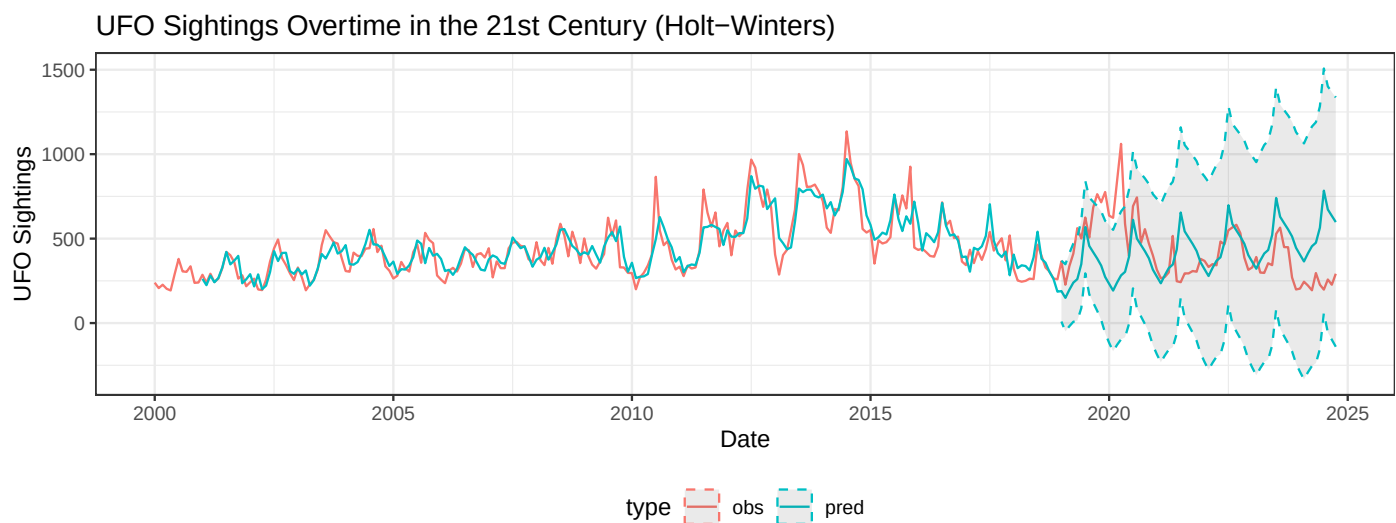


For the Holt-Winters model when looking at the ACF and PACF plots there doesn't seem to be significant residual autocorrelation. The ACF plot has a sinusoidal pattern in the magnitudes and there does seem to be lags where the magnitude of autocorrelation exceed the bounds of statistical significance (lags 4-7, 9, 12, 13, 18), however those magnitudes are not that large in magnitude when looking at the PACF plot which indicates that there isn't evidence against the independence assumption for the Holt-Winters model.

Results



The Auto.arima forecasting tool fit the data with an Autoregressive 1 Integrated 1 Moving Average 2 process with (0,1,2) seasonal components and a frequency of 12 ($ARIMA(1,1,2)(0,1,2)[12]$). Which ends up doing a solid job at fitting the actual observed data when looking at the graph, however forecasting the AR 1 process of the model puts a heavy emphasis on the time point directly prior in order to make a prediction of the current time point. As the UFO observations by month are trending downwards just prior to the cut off of Jan 2020 this results in the Auto.arima forecast to trend downwards which resulted in a very poor forecast.



For the Holt-Winters forecasting tool it fits the data with the parameters of alpha equal to 0.467319, beta of 0, and a gamma of 0.2530629. Essentially this means that the Holt-Winters model is putting a fair amount of weight on recent observations with an alpha of ~ 0.47 , however it is also considering the overall history of the time series when making its forecast as the alpha is not close to 1. A beta of 0 indicates that the HW forecast does not identify a trend component and a gamma of ~ 0.25 indicates

that HW identified a somewhat seasonal trend which is visible by the forecast repeating the same “cycle”. Holt-Winters does not experience the same decrease in the forecast as Auto.arima because of the weight it puts on the history of the time series which before the cut off of 2019 as a slight positive trend explaining why the forecast trends slightly positive. This leads to the Holt-Winters model having a better prediction, as seen in the sum of squared differences table below, since the data we tried to predict generally trends upwards and includes a large positive spike in the beginning.

Sum of Squared Differences For Each Forecasting Model

Auto.Arima	Holt_Winters
6590564	4091310

Discussion

Overall, both Auto.arima and Holt-Winters did a poor job of forecasting future sightings. Auto.arima forecasted a downward trend since it weighs more recent observations as more important. On the other hand Holt-Winters predicts an upward trend because it takes into account the overall trend of the data, which in this case, is mainly increasing. Holt-winters outperformed the auto.arima model when forecasting as quantified by sum of differences squared of the actual data to the forecast. The sum of squared differences of Holt-Winters was approximately 4 million and auto.arima’s was 6.6 million. If we were to improve our model, we would want to explore alternative datasets since we found that UFO sightings are not purely stochastic. Further analysis could be performed on other data, such as the total number of social media posts mentioning UFOs or how many times the word UFO was mentioned on certain platforms. We made an attempt to retrieve this data, but we were unsuccessful due to platforms like Twitter, Facebook, Instagram, and Reddit having pull limits and paywalls around their data. If we had more time to pursue this project, we would most likely move on to a free dataset on the internet of over one billion Reddit comments up until 2010 that was pulled from the internet before Reddit changed their terms and conditions. By including this extra data, our hope would be that we could better model some of the trends we were not able to explain using the stochastic methods we learned in class.

Code appendix

```
ufo <- read_csv("UFO data - Sheet1.csv")
ufo_clean <- ufo %>%
  mutate(Date = ym(Date)) %>%
  arrange(Date) %>%
  slice(601:n()-1)

ufo_clean %>%
  ggplot() +
  geom_line(aes(x = Date, y = `Number Sightings`), linewidth = 0.75) +
  geom_rect(xmin = as.numeric(as.Date("2020-01-01")),
            xmax = as.numeric(as.Date("2022-05-01")),
            ymin = 0, ymax = 2000, alpha = 0.005) +
  geom_vline(xintercept = as.numeric(as.Date("2019-07-01")), color="red",
             linetype = "dotted", linewidth=1)+
  geom_vline(xintercept = as.numeric(as.Date("2020-4-01")), color="red",
             linetype = "dotted", linewidth =1)+
  theme_bw() +
  labs(y = "Number of UFO Sightings",
       title = "Monthly UFO Sightings Since 2000")

ufo_ts <- ts(
  ufo_clean$`Number Sightings`,
  start = c(2000, 1),
  frequency = 12
)
boxplot(ufo_ts ~ cycle(ufo_ts), ylab = "UFO Sightings", xlab = "Month",
        main = "UFO Sightings by Month")

ufo_clean_red <- ufo_clean |>
  dplyr::slice(1:228)
ufo_ts <- ts(
  ufo_clean$`Number Sightings`,
  start = c(2000, 1),
  frequency = 12
)
ufo_red <- window(
  ufo_ts,
  start = c(2000,1),
  end = c(2018,12)
)
auto_ts_full<-auto.arima(ufo_ts)
auto_ts<-auto.arima(ufo_red)
```



```
#summary(auto_ts)

hw <- HoltWinters(ufo_red, seasonal = "additive")

qqnorm(resid(auto_ts), main = "Normal Q-Q Plot for ARIMA Model")
qqline(resid(auto_ts))

qqnorm(resid(hw), main = "Normal Q-Q Plot for Holt-Winters Model")
qqline(resid(hw))

acf(resid(auto_ts))
pacf(resid(auto_ts))

acf(resid(hw))
pacf(resid(hw))

auto_forecast <- forecast(auto_ts, h = 70, level = 95)
obs_df_full <- ufo_clean %>%
  mutate(time = c(time(ufo_ts)), fitted = fitted(auto_ts_full)) %>%
  dplyr::select(time, `Number Sightings`, fitted) %>%
  pivot_longer(-time)
obs_df <- ufo_clean_red %>%
  mutate(time = c(time(ufo_red)), fitted = fitted(auto_ts)) %>%
  dplyr::select(time, `Number Sightings`, fitted) %>%
  pivot_longer(-time)
forecast_df <- tibble(
  time = c(time(auto_forecast$mean)),
  value = c(auto_forecast$mean),
  lwr = c(auto_forecast$lower),
  upr = c(auto_forecast$upper),
  name = "forecast"
)
obs_df_avgs <- obs_df_full |>
  dplyr::filter(name == "Number Sightings")
obs_df_avg <- obs_df |>
  dplyr::filter(name == "Number Sightings")
obs_df_fit <- obs_df |>
  dplyr::filter(name == "fitted")
ggplot() +
  geom_line(data = obs_df_avgs, aes(x = time, y = value, col = name)) +
  geom_line(data = obs_df_fit, aes(x = time, y = value, col = name),
    linetype = 6) +
  geom_line(data = forecast_df, aes(x = time, y = value, col = name)) +
  geom_ribbon(data = forecast_df, aes(x = time, ymin = lwr, ymax = upr),
```

```

      alpha = .15) +
labs(title = "UFO Sightings Overtime in the 21st Century (Auto.ARIMA)") +
labs(x = "Date", y = "UFO Sightings") +
theme_bw() +
theme(legend.position = "bottom", legend.direction = "horizontal")

hw_pred <- predict(hw, n.ahead = 70, prediction.interval = TRUE)
bind_rows(
  tibble(
    time = time(ufo_ts),
    val = c(ufo_ts)) %>%
    mutate(type = "obs"),
  bind_rows(tibble(
    time = time(hw$fitted[,1]),
    val = c(hw$fitted[,1]),
    tibble(time = time(hw_pred),
            val = c(hw_pred[,1]),
            lwr = c(hw_pred[,3]),
            upr = c(hw_pred[,2])) %>%
      mutate(type = "pred")) %>%
    ggplot(aes(x = time, y = val, col = type)) +
    geom_line() +
    geom_ribbon(aes(ymin = lwr, ymax = upr),
      linetype = 2,
      alpha = 0.1) +
labs(title = "UFO Sightings Overtime in the 21st Century (Holt-Winters)") +
labs(x = "Date", y = "UFO Sightings") +
theme_bw() +
theme(legend.position = "bottom", legend.direction = "horizontal")

diffs <- window(ufo_ts, start = c(2019, 1), end = c(2024, 10)) - hw_pred[,1]
Holt_Winters <- sum(diffs^2)

diffs1 <- as.data.frame(window(ufo_ts, start = c(2019, 1),
                              end = c(2024, 10))) - forecast_df[,2]
Auto.Arima <- sum(diffs1^2)

sums_diffs <- tibble(Auto.Arima, Holt_Winters)
#sums_diffs %>%
#  kbl(digits = 5, format = "latex", position = "h!",
#    caption = "Sum of Squared Differences For Each Forecasting Model") %>%
#  row_spec(0, bold=TRUE) %>%
#  kable_classic()

```